Lesson 11 Teacher's Key: Wave After Wave

GDevelop - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Splitting one event (Lesson 6's single win-check) into two mutually exclusive events (full-win vs. advance-wave), each guarded by an opposite comparison on the same Wave variable - GDevelop's event-based way of expressing an if/else.

Level 1 - Full Worked Solution

- Wave and TotalWaves are new scene variables, parallel to Score and Lives.
- The full-win event and the advance-wave event share the same first condition (EnemiesRemaining <= 0) but have opposite second conditions (Wave >= TotalWaves vs. Wave < TotalWaves) - together, they cover every possibility exactly once, with no overlap.
- The advance-wave event's spawn action is a direct copy of the Setup group's spawn action - GDevelop doesn't require extracting a callable function for this to work, though the optional Level 3 Functions idea shows students how they could.

Why This Structure

- GDevelop's event-condition model expresses 'if/else' as two separate, oppositely-guarded events rather than a single event with a branching action list - a genuinely different (and equally valid) way of achieving the same logical structure other engines build with code.
- Keeping WinText/LoseText reserved for the round's true end (not wave transitions) continues the established discipline from Lesson 6 and Lesson 9.

Level 2 / Level 3 Talking Points

- Ask students to explain, in their own words, why the two new events' Wave conditions need to be exact opposites (>= vs. <) - if they overlapped or left a gap, some wave-clear moments could trigger both, neither, or the wrong one.
- For students ready for a stretch: point out the Level 3 idea about extracting the spawn action into a genuine GDevelop Function (Project > Functions) - not required, but a natural bridge toward the 'named reusable routine' pattern every other engine uses this week.

Common Student Mistakes

- Only one of the two new events' Wave conditions actually added (e.g., forgetting Wave < TotalWaves on the advance event) - can cause both events to fire on the same frame, or neither.
- The advance-wave event's spawn action missing or referencing the wrong count.
- Forgetting Wave = 1 in the Setup group, leaving it unreset after a restart.